

Applied NLP

DATA 641

Lecture 3 — NLP Pipeline; Deployment and monitoring

*Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems 1st Edition, Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, O' Reilly.

Deployment

- In most practical application scenarios, the NLP module we are implementing is a part of a larger system.
- Once we are happy with one final solution, it needs to be deployed in a production environment as a part of a larger system.
- **Deployment** entails plugging the NLP module into the broader system. It may also involve making sure input and output data pipelines are in order, as well as making sure our NLP module is scalable under heavy load.

Monitoring and model updating

Monitoring for NLP projects and models has to be handled differently than a regular engineering project, as we need to ensure that the outputs produced by our models daily make sense.

Project attribute	Decision paths	Examples
More training data is generated post-deployment.	Once deployed, extracted signals can be used to automatically improve the model. Can also try online learning to train the model automatically on a daily basis.	Abuse-detection systems where users flag data.
Training data is not generated post-deployment.	Manual labeling could be done to improve evaluation and the models. Ideally, each new model has to be manually built and evaluated.	A subset of a larger NLP pipeline with no direct feedback.
Low model latency is required, or model has to be online with near-real-time response.	Need to use models that can be inferred quickly. Another option is to create memoization strategies like caching or have substantially bigger computing power.	Systems that need to respond right away, like any chatbot or an emergency tracking system.
Low model latency is not required, or model can be run in an offline fashion.	Can use more advanced and slower models. This can also help in optimizing costs where feasible.	Systems that can be run on a batch process, like retail product catalog analysis.